Step 1: we list all the transactions

T1: {Bread, Milk, Chips, Mustard}
T2: {Beer, Diaper, Bread, Eggs}
T3: {Beer, Coke, Diaper, Milk}
T4: {Beer, Bread, Diaper, Milk, Chips}
T5: {Coke, Bread, Diaper, Milk}
T6: {Beer, Bread, Diaper, Milk, Mustard}
T7: {Coke, Bread, Diaper, Milk}

Step 2: we find frequent 1-itemsets (L1)

Count each item

- Bread: 6
- Milk: 6
- Diaper: 6
- Beer: 4
- Coke: 3
- Chips: 2
- Mustard: 2
- Eggs: 1

Frequent 1-itemsets (count ≥ 3)

L1 =

- {Bread} (6)
- {Milk} (6)
- {Diaper} (6)
- {Beer} (4)
- {Coke} (3)

Step 3: we generate candidate 2-itemsets (C2) from L1, then count L2

All pairs from {Bread, Milk, Diaper, Beer, Coke} and their counts:

- {Bread, Milk}: 5
- {Bread, Diaper}: 5
- {Milk, Diaper}: 5
- {Beer, Diaper}: 4
- {Bread, Beer}: 3

- {Milk, Beer}: 3
- {Milk, Coke}: 3
- {Diaper, Coke}: 3
- {Bread, Coke}: 2
- {Beer, Coke}: 1

Frequent 2-itemsets (L2)

L2 =

- {Bread, Milk} (5)
- {Bread, Diaper} (5)
- {Milk, Diaper} (5)
- {Beer, Diaper} (4)
- {Bread, Beer} (3)
- {Milk, Beer} (3)
- {Milk, Coke} (3)
- {Diaper, Coke} (3)

Step 4: we generate candidate 3-itemsets (C3) using Apriori pruning, then count L3

Apriori rule:

A 3-itemset is a valid candidate only if all its 2-item subsets are frequent (must be in L2).

Valid candidates and counts:

- {Bread, Milk, Beer}: 2 (this is not frequent)
- {Bread, Milk, Diaper}: 4
- {Bread, Beer, Diaper}: 3
- {Milk, Beer, Diaper}: 3
- {Milk, Diaper, Coke}: 3

Frequent 3-itemsets (L3)

L3 =

- {Bread, Milk, Diaper} (4)
- {Bread, Beer, Diaper} (3)
- {Milk, Beer, Diaper} (3)
- {Milk, Diaper, Coke} (3)

Step 5: and then try 4-itemsets (C4)

- {Bread, Milk, Beer} = 2 (not frequent)

So {Bread, Milk, Beer, Diaper} is pruned, and there are no frequent 4-itemsets.

Here we stop.

Therefore the frequent itemsets are”

L1: {Bread}, {Milk}, {Diaper}, {Beer}, {Coke}

L2: {Bread, Milk}, {Bread, Diaper}, {Milk, Diaper}, {Beer, Diaper},
{Bread, Beer}, {Milk, Beer}, {Milk, Coke}, {Diaper, Coke}

L3: {Bread, Milk, Diaper}, {Bread, Beer, Diaper}, {Milk, Beer, Diaper}, {Milk, Diaper, Coke}